NetInfra -
A Framework for Expressing Network Infrastructure as Code

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Background

- Kea/Bind 9
- Configuration management
- NetInfra as a bridge

ISC-Kea Logo: https://www.isc.org/images/isclogos/kea-imageonly-250.png // ISC-Bind9 Logo: https://www.isc.org/images/isclogos/Bind_9_Mark_ISC_Blue.png

Solution Design

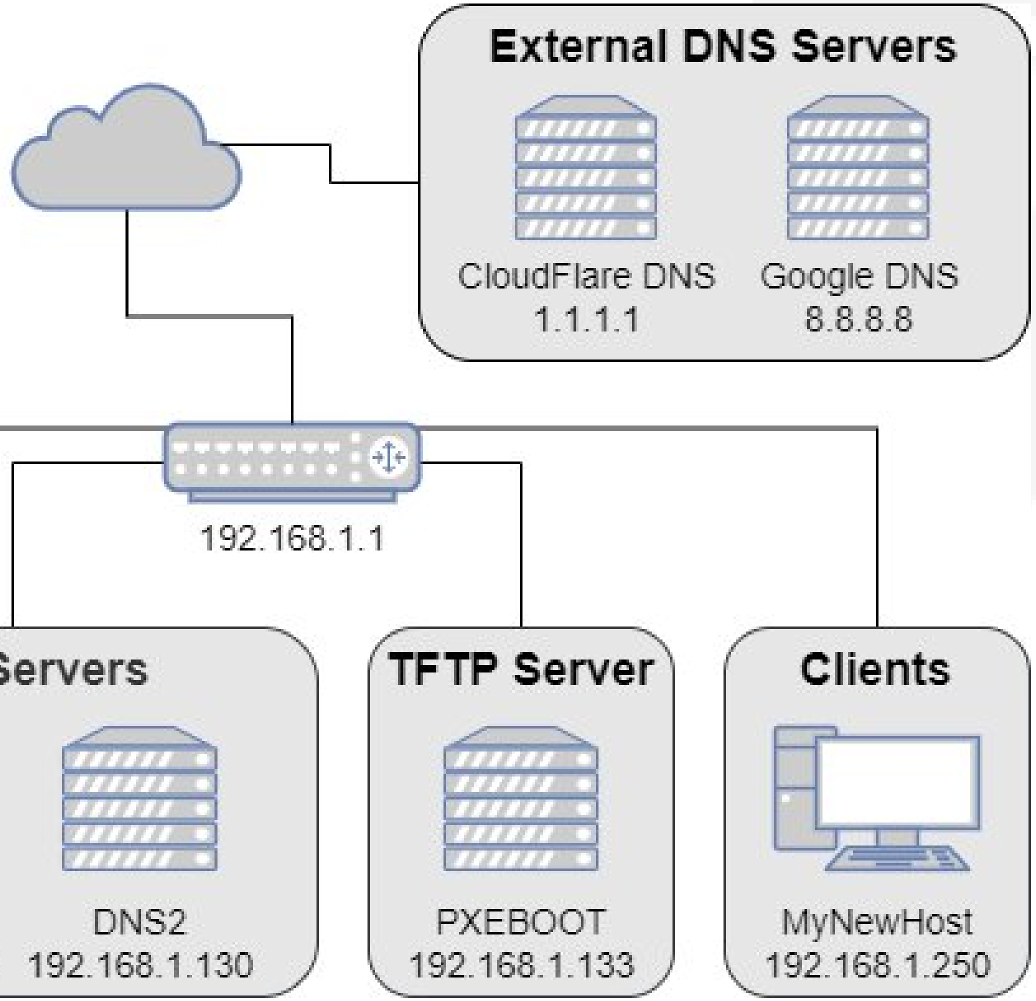
Goals

- Single File
- DHCP & DNS Servers, Subnets, & Zones
- All Hosts
- Ansible Compatible
- ISC-Kea & ISC-Bind naming conventions

Benefits

- Human Readable
- Single Source of Truth
- Single Structure to Learn
- Error Checking
- Easily Modified & Extendable

Example Network Layout



Server Declarations

DHCP Servers

```yaml
dhcp_servers:
  - name: dhcp1
    ip: 192.168.1.131
    role: primary
  - name: dhcp2
    ip: 192.168.1.132
    role: secondary
next_server: 192.168.1.133
```

DNS Servers

```yaml
dns_servers:
  - name: dns1.example.com
    ip: 192.168.1.129
    type: primary
  - name: dns2.example.com
    ip: 192.168.1.130
    type: secondary

external_dns_servers:
  - 1.1.1.1
  - 8.8.8.8
```

DHCP Subnet Declarations

```yaml
subnets:
  - id: 100
    cidr: 192.168.1.0/24
    dns_servers: "192.168.1.129, 192.168.1.130"
    subnet_mask: 255.255.255.0
    routers: 192.168.1.1
    dhcp_options:
      - interface_mtu: 1500
    pools:
      - "192.168.1.64/28"
      - "192.168.1.79-192.168.1.127"
  - id: 101
    cidr:  10.10.0.0/16
    dns_servers: "192.168.1.129, 192.168.1.130"
    subnet_mask: 255.255.0.0
    routers: 10.10.0.1
    required_client_classes:
      - "custom-example"
```

DNS Zone Declarations

Forward Zones

```
zones:
  - name: example.com
    base_serial: 100000
  - name: example.org
    file: example.com
  - name: example.net
    ttl: 2h
    refresh: 2h
    retry: 1h
    expire: 2W
    min_ttl: 5m
```

Reverse Zones

```
rzones:
  - name: 1.168.192.in-addr.arpa
    regex: "192.168.1."
    base_serial: 100000
  - name: 10.10.in-addr.arpa
    regex: "10.10."
    context: "2"
    base_serial: 100000
    ttl: 2h
    refresh: 2h
    retry: 1h
    expire: 2W
    min_ttl: 5m
```

Host Declarations

```yaml
hosts:
  - name: mynewhost
    records:
      - zonename: example.com
        target: 192.168.1.250
        mac: 00:11:22:aa:bb:cc
        comment: "Example A Record"
      - zonename: example.org
        source: oldhostname
        ttl: 60
        target: mynewhost.example.com
        rrtype: CNAME
        comment: "Example CNAME"
```

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Implementation

NetInfra

Roles:

- kea
- bind9

Templates:

- kea-dhcp4.conf.j2
- forward.zone.j2
- reverse.zone.j2

Ansible Logo: https://commons.wikimedia.org/wiki/File:Ansible_Logo.png // Jinja Logo: https://jinja.palletsprojects.com/en/3.1.x/_images/jinja-logo.png

NetInfra Workflow

1. New host added to network.
2. Admin adds host entry to NetInfra.yml
3. Push changes to version control.
4. Deploy changes via configuration management.
5. DHCP reservation created automatically.
6. DNS records created automatically.

Lessons Learned

File Design

Incompatible Syntaxes

Deployment

Future Work

Dynamic configuration files	Auto-discovery	Large Groupings
GUI	IPv6	Other DHCP/DNS packages

Thank-you!

Any Questions?

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Connect With Us!

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